21. Show that if $\alpha = 1/n$ where n is a positive integer, then the principal value of z^α is the same as the principal nth root of z.

22. (**a**) Show that if α is a rational number (that is, $\alpha = m/n$ where m and n are integers with no common factor), then z^α is finite-valued. In particular, show that there are n distinct values of $z^{m/n}$.

(**b**) Show that if α is an irrational number (that is, not a rational number) or a complex number, then z^α is infinite-valued.

23. Suppose that α is a real number. For a fixed nonzero complex number z_0, show that all values of z_0^α lie on the same circle in the complex plane.

24. A useful property of real numbers is $x^a y^a = (xy)^a$.

(**a**) Does the property $z^\alpha w^\alpha = (zw)^\alpha$ hold for complex powers?

(**b**) Does the property $z^\alpha w^\alpha = (zw)^\alpha$ hold for the principal value of a complex power?

Computer Lab Assignments

Most CASs have a built in function to find the principal value of a complex power. In *Mathematica*, the syntax **(a + b I)^(c + d I)** is used to accomplish this. To find a numerical approximation of this value in *Mathematica* you enter the expression **N[(a + b I)^(c + d I)]**. For example, *Mathematica* indicates that **N[(1 + 2 I)^(3 + 2 I)]** is approximately $0.2647 - 1.1922i$.

In Problems 25–30, use a CAS to find the principal value of the given complex power.

25. $(1 - 5i)^i$

26. 5^{5-2i}

27. $(2 - i)^{3+2i}$

28. $(1 - 4i)^{1+3i}$

29. $(1 + i)^{(1+i)^{1+i}}$

30. $(1 - 3i)^{1/4}$

4.3 Trigonometric and Hyperbolic Functions

INTRODUCTION In this section we define the complex trigonometric and hyperbolic functions. As with the complex functions e^z and Ln z defined in a previous section, these functions will agree with their real counterparts for real input. In addition, we will show that the complex trigonometric and hyperbolic functions have the same derivatives and satisfy many of the same identities as the real trigonometric and hyperbolic functions.

4.3.1 Complex Trigonometric Functions

If x is a real variable, then it follows from Definition 4.1.1 that:

$$e^{ix} = \cos x + i \sin x \quad \text{and} \quad e^{-ix} = \cos x - i \sin x. \tag{1}$$

By adding these equations and simplifying, we obtain an equation that relates the real cosine function with the complex exponential function:

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}. \tag{2}$$

In a similar manner, if we subtract the two equations in (1), then we obtain an expression for the real sine function:

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}. \tag{3}$$

The formulas for the *real* cosine and sine functions given in (2) and (3) can be used to define the *complex* sine and cosine functions. Namely, we define these complex trigonometric functions by replacing the real variable x with the complex variable z in (2) and (3).

DEFINITION 4.3.1 Complex Sine and Cosine Functions

The complex **sine** and **cosine** functions are defined by:

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad \text{and} \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}. \tag{4}$$

A review of real trigonometric functions can be found in Section 4.3 of the student study guide. ▶

It follows from (2) and (3) that the complex sine and cosine functions defined by (4) agree with the real sine and cosine functions for real input. We next define the complex tangent, cotangent, secant, and cosecant functions using the complex sine and cosine:

$$\tan z = \frac{\sin z}{\cos z}, \quad \cot z = \frac{\cos z}{\sin z}, \quad \sec z = \frac{1}{\cos z}, \quad \text{and} \quad \csc z = \frac{1}{\sin z}. \tag{5}$$

These functions also agree with their real counterparts for real input.

EXAMPLE 1 Values of Complex Trigonometric Functions

In each part, express the value of the given trigonometric function in the form $a + ib$.

(**a**) $\cos i$ (**b**) $\sin(2 + i)$ (**c**) $\tan(\pi - 2i)$

Solution For each expression we apply the appropriate formula from (4) or (5) and simplify.

(**a**) By (4),

$$\cos i = \frac{e^{i \cdot i} + e^{-i \cdot i}}{2} = \frac{e^{-1} + e}{2} \approx 1.5431.$$

(**b**) By (4),

$$\begin{aligned}
\sin(2 + i) &= \frac{e^{i(2+i)} - e^{-i(2+i)}}{2i} \\
&= \frac{e^{-1+2i} - e^{1-2i}}{2i} \\
&= \frac{e^{-1}(\cos 2 + i \sin 2) - e(\cos(-2) + i \sin(-2))}{2i} \\
&= \left(\frac{e + e^{-1}}{2}\right) \sin 2 + i \left(\frac{e - e^{-1}}{2}\right) \cos 2 \\
&\approx 1.4031 - 0.4891i.
\end{aligned}$$

(**c**) By the first entry in (5) together with (4) we have:

$$\begin{aligned}
\tan(\pi - 2i) &= \frac{\left(e^{i(\pi-2i)} - e^{-i(\pi-2i)}\right)/2i}{\left(e^{i(\pi-2i)} + e^{-i(\pi-2i)}\right)/2} = \frac{e^{i(\pi-2i)} - e^{-i(\pi-2i)}}{\left(e^{i(\pi-2i)} + e^{-i(\pi-2i)}\right) i} \\
&= -\frac{e^2 - e^{-2}}{e^2 + e^{-2}} i \approx -0.9640i.
\end{aligned}$$

☰

Identities Most of the familiar identities for real trigonometric functions hold for the complex trigonometric functions. This follows from Definition 4.3.1 and properties of the complex exponential function. We now list some of the more useful of the trigonometric identities. Each of the results in (6)–(10) is identical to its real analogue.

$$\sin(-z) = -\sin z \quad \cos(-z) = \cos z \tag{6}$$

$$\cos^2 z + \sin^2 z = 1 \tag{7}$$

$$\sin(z_1 \pm z_2) = \sin z_1 \cos z_2 \pm \cos z_1 \sin z_2 \tag{8}$$

$$\cos(z_1 \pm z_2) = \cos z_1 \cos z_2 \mp \sin z_1 \sin z_2 \tag{9}$$

Observe that the double-angle formulas:

$$\sin 2z = 2 \sin z \cos z \quad \cos 2z = \cos^2 z - \sin^2 z \tag{10}$$

follow directly from (8) and (9).

We will verify only identity (7). The other identities follow in a similar manner. See Problems 13 and 14 in Exercises 4.3. In order to verify (7), we note that by (4) and properties of the complex exponential function from Theorem 4.1.2, we have

$$\cos^2 z = \left(\frac{e^{iz} + e^{-iz}}{2}\right)^2 = \frac{e^{2iz} + 2 + e^{-2iz}}{4},$$

and

$$\sin^2 z = \left(\frac{e^{iz} - e^{-iz}}{2i}\right)^2 = -\frac{e^{2iz} - 2 + e^{-2iz}}{4}.$$

Therefore,

$$\cos^2 z + \sin^2 z = \frac{e^{2iz} + 2 + e^{-2iz} - e^{2iz} + 2 - e^{-2iz}}{4} = 1.$$

Note ▶ It is important to recognize that some properties of the real trigonometric functions are *not* satisfied by their complex counterparts. For example, $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all real x, but, from Example 1 we have $|\cos i| > 1$ and $|\sin(2 + i)| > 1$ since $|\cos i| \approx 1.5431$ and $|\sin(2 + i)| \approx 1.4859$, so these inequalities, in general, are not satisfied for complex input.

Periodicity In Section 4.1 we saw that the complex exponential function is periodic with a pure imaginary period of $2\pi i$. That is, $e^{z+2\pi i} = e^z$ for all complex z. Replacing z with iz in this equation we obtain $e^{iz+2\pi i} = e^{i(z+2\pi)} = e^{iz}$. Thus, e^{iz} is a periodic function with real period 2π. Similarly, we can show that $e^{-i(z+2\pi)} = e^{-iz}$ and so e^{-iz} is also a periodic function with a real period of 2π. Now from Definition 4.3.1 it follows that:

$$\sin(z + 2\pi) = \frac{e^{i(z+2\pi)} - e^{-i(z+2\pi)}}{2i} = \frac{e^{iz} - e^{-iz}}{2i} = \sin z.$$

A similar statement also holds for the complex cosine function. In summary, we have:

$$\sin(z + 2\pi) = \sin z \quad \text{and} \quad \cos(z + 2\pi) = \cos z \tag{11}$$

for all z. Put another way, (11) shows that the complex sine and cosine are periodic functions with a real period of 2π. The periodicity of the secant and cosecant functions follows immediately from (11) and (5). The identities $\sin(z + \pi) = -\sin z$ and $\cos(z + \pi) = -\cos z$ can be used to show that the complex tangent and cotangent are periodic with a real period of π. See Problems 51 and 52 in Exercises 4.3.

Trigonometric Equations We now turn our attention to solving simple trigonometric equations. Because the complex sine and cosine functions are periodic, there are always infinitely many solutions to equations of the form $\sin z = w$ or $\cos z = w$. One approach to solving such equations is to use Definition 4.3.1 in conjunction with the quadratic formula. We demonstrate this method in the following example.

EXAMPLE 2 Solving Trigonometric Equations

Find all solutions to the equation $\sin z = 5$.

Solution By Definition 4.3.1, the equation $\sin z = 5$ is equivalent to the equation

$$\frac{e^{iz} - e^{-iz}}{2i} = 5.$$

By multiplying this equation by e^{iz} and simplifying we obtain

$$e^{2iz} - 10ie^{iz} - 1 = 0.$$

This equation is quadratic in e^{iz}. That is,

$$e^{2iz} - 10ie^{iz} - 1 = \left(e^{iz}\right)^2 - 10i\left(e^{iz}\right) - 1 = 0.$$

Thus, it follows from the quadratic formula (3) of Section 1.6 that the solutions of $e^{2iz} - 10ie^{iz} - 1 = 0$ are given by

$$e^{iz} = \frac{10i + (-96)^{1/2}}{2} = 5i \pm 2\sqrt{6}i = \left(5 \pm 2\sqrt{6}\right)i. \tag{12}$$

In order to find the values of z satisfying (12), we solve the two exponential equations in (12) using the complex logarithm. If $e^{iz} = \left(5 + 2\sqrt{6}\right)i$, then $iz = \ln\left(5i + 2\sqrt{6}i\right)$ or $z = -i\ln\left[\left(5 + 2\sqrt{6}\right)i\right]$. Because $\left(5 + 2\sqrt{6}\right)i$ is a pure imaginary number and $5 + 2\sqrt{6} > 0$, we have $\arg\left[\left(5 + 2\sqrt{6}\right)i\right] = \frac{1}{2}\pi + 2n\pi$. Thus,

$$z = -i\log\left[\left(5 + 2\sqrt{6}\right)i\right] = -i\left[\log_e\left(5 + 2\sqrt{6}\right) + i\left(\frac{\pi}{2} + 2n\pi\right)\right]$$

or

$$z = \frac{(4n+1)\pi}{2} - i\log_e\left(5 + 2\sqrt{6}\right) \tag{13}$$

for $n = 0, \pm1, \pm2, \ldots$. In a similar manner, we find that if $e^{iz} = \left(5 - 2\sqrt{6}\right)i$, then $z = -i\ln\left[\left(5 - 2\sqrt{6}\right)i\right]$. Since $\left(5 - 2\sqrt{6}\right)i$ is a pure imaginary number and $5 - 2\sqrt{6} > 0$, it has an argument of $\pi/2$, and so:

$$z = -i\log\left[\left(5 - 2\sqrt{6}\right)i\right] = -i\left[\log_e\left(5 - 2\sqrt{6}\right) + i\left(\frac{\pi}{2} + 2n\pi\right)\right]$$

or

$$z = \frac{(4n+1)\pi}{2} - i\log_e\left(5 - 2\sqrt{6}\right) \tag{14}$$

for $n = 0, \pm1, \pm2, \ldots$. Therefore, we have shown that if $\sin z = 5$, then z is one of the values given in (13) or (14). ☰

Modulus The modulus of a complex trigonometric function can also be helpful in solving trigonometric equations. To find a formula in terms of x and y for the modulus of the sine and cosine functions, we first express these functions in terms of

their real and imaginary parts. If we replace the symbol z with the symbol $x + iy$ in the expression for $\sin z$ in (4), then we obtain:

$$\sin z = \frac{e^{-y+ix} - e^{y-ix}}{2i} = \frac{e^{-y}(\cos x + i \sin x) - e^{y}(\cos x - i \sin x)}{2i}$$
$$= \sin x \left(\frac{e^{y} + e^{-y}}{2}\right) + i \cos x \left(\frac{e^{y} - e^{-y}}{2}\right). \tag{15}$$

Since the real hyperbolic sine and cosine functions are defined by $\sinh y = \dfrac{e^{y} - e^{-y}}{2}$ and $\cosh y = \dfrac{e^{y} + e^{-y}}{2}$ we can rewrite (15) as

$$\sin z = \sin x \cosh y + i \cos x \sinh y. \tag{16}$$

A similar computation enables us to express the complex cosine function in terms of its real and imaginary parts as:

$$\cos z = \cos x \cosh y - i \sin x \sinh y. \tag{17}$$

We now use (16) and (17) to derive formulas for the modulus of the complex sine and cosine functions. From (16) we have:

$$|\sin z| = \sqrt{\sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y}.$$

This formula can be simplified using the identities $\cos^2 x + \sin^2 x = 1$ and $\cosh^2 y = 1 + \sinh^2 y$ for the real trigonometric and hyperbolic functions:

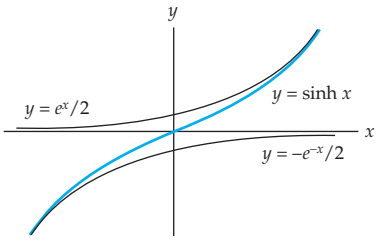
$$|\sin z| = \sqrt{\sin^2 x \left(1 + \sinh^2 y\right) + \cos^2 x \sinh^2 y}$$
$$= \sqrt{\sin^2 x + \left(\cos^2 x + \sin^2 x\right) \sinh^2 y},$$

or
$$|\sin z| = \sqrt{\sin^2 x + \sinh^2 y}. \tag{18}$$

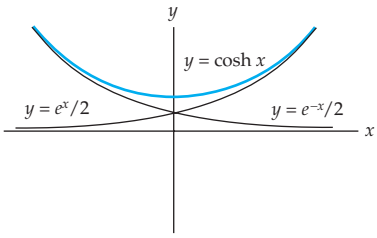
After a similar computation we obtain the following expression for the modulus of the complex cosine function:

$$|\cos z| = \sqrt{\cos^2 x + \sinh^2 y}. \tag{19}$$

You may recall from calculus that the real hyperbolic function $\sinh x$ is unbounded on the real line. See **FIGURE 4.3.1(a)**. As a result of this fact, the expressions in (18) and (19) can be made arbitrarily large by choosing y to be arbitrarily large. Thus, the complex sine and cosine functions are unbounded on the complex plane. That is, there does not exist a real constant M so that $|\sin z| < M$ for all z in $\mathbf{C}$, nor does there exist a real constant M so that $|\cos z| < M$ for all z in $\mathbf{C}$. This, of course, is quite different from the situation for the real sine and cosine functions for which $|\sin x| \leq 1$ and $|\cos x| \leq 1$ for all real x.



(a) $y = \sinh x$



(b) $y = \cosh x$

FIGURE 4.3.1 The real hyperbolic functions

Zeros The formulas derived for the modulus of the complex sine and cosine are helpful in determining the zeros of these functions. Recall that the zeros of the real sine function occur at integer multiples of π, and that the zeros of the real cosine function occur at odd integer multiples of $\pi/2$. Since the complex sine and cosine functions agree with their real counterparts for real input, it follows that these zeros of the *real* sine and cosine functions are also zeros of the *complex* sine and cosine functions. It is a natural question to ask whether the complex sine and cosine

functions have any additional zeros in the complex plane. One way of answering this question is by solving the equations $\sin z = 0$ and $\cos z = 0$ in the manner presented in Example 2. A different method involves recognizing that a complex number is equal to 0 if and only if its modulus is 0. Thus, solving the equation $\sin z = 0$ is equivalent to solving the equation $|\sin z| = 0$. Using (18) we see that if $|\sin z| = 0$, then $\sqrt{\sin^2 x + \sinh^2 y} = 0$, which is equivalent to:

$$\sin^2 x + \sinh^2 y = 0.$$

Since $\sin^2 x$ and $\sinh^2 y$ are both nonnegative real numbers, this last equation is satisfied if and only if $\sin x = 0$ and $\sinh y = 0$. As just noted, $\sin x = 0$ when $x = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$, and inspection of Figure 4.3.1(a) indicates that $\sinh y = 0$ only when $y = 0$. Therefore, the only solutions of the equation $\sin z = 0$ in the complex plane are the real numbers $z = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$. That is, the zeros of the complex sine function are the same as the zeros of the real sine functions; there are no additional zeros of sine in the complex plane. This is not the same as the situation for polynomial functions where there are often additional zeros in the complex plane.

In essentially the same manner we can show that the only zeros of the complex cosine function are the real numbers $z = (2n + 1)\pi/2$, $n = 0, \pm 1, \pm 2 \ldots$. See Problem 41 in Exercises 4.3. In summary we have:

All of the zeros of $\sin z$ and $\cos z$ are real. ▶

$$\sin z = 0 \text{ if and only if } z = n\pi, \tag{20}$$

and

$$\cos z = 0 \text{ if and only if } z = \frac{(2n+1)\pi}{2} \tag{21}$$

for $n = 0, \pm 1, \pm 2, \ldots$.

Analyticity The derivatives of the complex sine and cosine functions are found using the chain rule (6) in Section 3.2. For the complex sine function we have:

$$\frac{d}{dz}\sin z = \frac{d}{dz}\left(\frac{e^{iz} - e^{-iz}}{2i}\right) = \frac{ie^{iz} + ie^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2},$$

or

$$\frac{d}{dz}\sin z = \cos z.$$

Since this derivative is defined for all complex z, $\sin z$ is an entire function. In a similar manner, we find

$$\frac{d}{dz}\cos z = -\sin z.$$

The derivatives of $\sin z$ and $\cos z$ can then be used to show that derivatives of all of the complex trigonometric functions are the same as derivatives of the real trigonometric functions. The derivatives of the six complex trigonometric functions are summarized in the following.

Derivatives of Complex Trigonometric Functions

$$\frac{d}{dz}\sin z = \cos z \qquad \frac{d}{dz}\cos z = -\sin z$$
$$\frac{d}{dz}\tan z = \sec^2 z \qquad \frac{d}{dz}\cot z = -\csc^2 z$$
$$\frac{d}{dz}\sec z = \sec z \tan z \qquad \frac{d}{dz}\csc z = -\csc z \cot z$$

The sine and cosine functions are entire, but the tangent, cotangent, secant, and cosecant functions are only analytic at those points where the denominator is nonzero. From (20) and (21), it then follows that the tangent and secant functions have singularities at $z = (2n + 1)\pi/2$ for $n = 0, \pm 1, \pm 2 \ldots$, whereas the cotangent and cosecant functions have singularities at $z = n\pi$ for $n = 0, \pm 1, \pm 2 \ldots$.

Trigonometric Mapping We will now discuss the complex mapping $w = \sin z$ of the z-plane onto the w-plane. Because $\sin z$ is periodic with a real period of 2π, this function takes on all values in any infinite vertical strip $x_0 < x \leq x_0 + 2\pi$, $-\infty < y < \infty$. In a manner similar to that used to study the exponential mapping $w = e^z$, this allows us to study the mapping $w = \sin z$ on the entire complex plane by analyzing it on any one of these strips. Consider, say, the strip $-\pi < x \leq \pi$, $-\infty < y < \infty$. Before we examine the complex mapping $w = \sin z$ on this strip, observe that $\sin z$ is not one-to-one on this region. For example, $z_1 = 0$ and $z_2 = \pi$ are in this region and $\sin 0 = \sin \pi = 0$. From the identity $\sin(-z + \pi) = \sin z$ it follows that the image of the strip $-\pi < x \leq -\pi/2$, $-\infty < y < \infty$, is the same as the image of the strip $\pi/2 < x \leq \pi$, $-\infty < y < \infty$, under $w = \sin z$. Therefore, we need only consider the mapping $w = \sin z$ on the region $-\pi/2 \leq x \leq \pi/2$, $-\infty < y < \infty$, to gain an understanding of this mapping on the entire z-plane. In Problem 45 of Exercises 4.3 you will be asked to show that the complex sine function is one-to-one on the domain $-\pi/2 < x < \pi/2$, $-\infty < y < \infty$.

EXAMPLE 3 The Mapping $w = \sin z$

Describe the image of the region $-\pi/2 \leq x \leq \pi/2$, $-\infty < y < \infty$, under the complex mapping $w = \sin z$.

Solution Similar to the discussion on pages 154–155 of Section 4.1, one approach to this problem is to determine the image of vertical lines $x = a$ with $-\pi/2 \leq a \leq \pi/2$ under $w = \sin z$. Assume for the moment that $a \neq -\pi/2$, 0, or $\pi/2$. From (16) the image of the vertical line $x = a$ under $w = \sin z$ is given by:

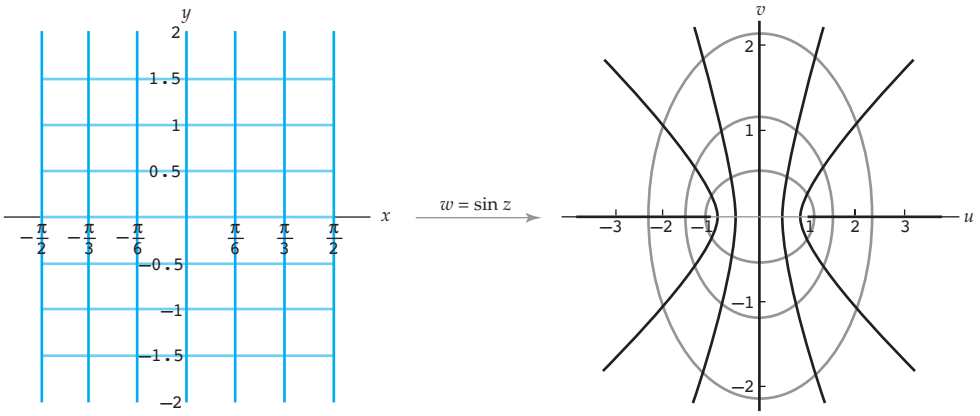
$$u = \sin a \cosh y, \quad v = \cos a \sinh y, \quad -\infty < y < \infty. \tag{22}$$

We will eliminate the variable y in (22) to obtain a single Cartesian equation relating u and v. Since $-\pi/2 < a < \pi/2$ and $a \neq 0$, it follows that $\sin a \neq 0$ and $\cos a \neq 0$, and so from (22) we obtain $\cosh y = \dfrac{u}{\sin a}$ and $\sinh y = \dfrac{v}{\cos a}$. The identity $\cosh^2 y - \sinh^2 y = 1$ for real hyperbolic functions then gives the following equation:

$$\left(\frac{u}{\sin a}\right)^2 - \left(\frac{v}{\cos a}\right)^2 = 1. \tag{23}$$

The Cartesian equation in (23) is a hyperbola with vertices at $(\pm \sin a,\ 0)$ and slant asymptotes $v = \pm\left(\dfrac{\cos a}{\sin a}\right) u$. Because the point $(a,\ 0)$ is on the line $x = a$, the point $(\sin a,\ 0)$ must be on the image of the line. Therefore, the image of the vertical line $x = a$ with $-\pi/2 < a < \pi/2$ and $a \neq 0$ under $w = \sin z$ is the branch* of the hyperbola (23) that contains the point $(\sin a, 0)$. Because $\sin(-z) = -\sin z$ for all z, it also follows that the image of the line $x = -a$ is branch of the hyperbola (23) containing the point $(-\sin a,\ 0)$. We illustrate this mapping property of $w = \sin z$ in FIGURE 4.3.2, where the vertical lines shown in color in Figure 4.3.2(a) are mapped onto the hyperbolas shown in black in Figure 4.3.2(b). The line $x = \pi/3$ is mapped

*Do not confuse this term with "branch of a multiple-valued function."



(a) The region $-\pi/2 \le x \le \pi/2$ **(b)** Image of the region in (a)

FIGURE 4.3.2 The mapping $w = \sin z$

onto the branch of the hyperbola containing the point $\left(\frac{1}{2}\sqrt{3}, 0\right)$ and the line $x = \pi/6$ is mapped onto the branch containing the point $\left(\frac{1}{2}, 0\right)$. Similarly, the line $x = -\pi/3$ is mapped onto the branch containing the point $\left(-\frac{1}{2}\sqrt{3}, 0\right)$ and the line $x = -\pi/6$ is mapped onto the branch containing the point $\left(-\frac{1}{2}, 0\right)$.

The images of the lines $x = -\pi/2$, $x = \pi/2$ and $x = 0$ cannot be found from (23). However, from (22) we see that the image of the line $x = -\pi/2$ is the set of points $u \le -1$ on the negative real axis, that the image of the line $x = \pi/2$ is the set of points $u \ge 1$ on the positive real axis, and that the image of the line $x = 0$ is the imaginary axis $u = 0$. See Figure 4.3.2. In summary, we have shown that the image of the infinite vertical strip $-\pi/2 \le x \le \pi/2$, $-\infty < y < \infty$, under $w = \sin z$, is the entire w-plane. ☰

In Example 3 the image could also be found using horizontal line segments $y = b$, $-\pi/2 \le x \le \pi/2$, instead of vertical lines. In this case, the images are given by:

$$u = \sin x \cosh b, \quad v = \cos x \sinh b, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

When $b \neq 0$, this set is also given by the Cartesian equation:

$$\left(\frac{u}{\cosh b}\right)^2 + \left(\frac{v}{\sinh b}\right)^2 = 1, \tag{24}$$

which is an ellipse with u-intercepts at $(\pm \cosh b,\ 0)$ and v-intercepts at $(0,\ \pm \sinh b)$. If $b > 0$, then the image of the line segment $y = b$ is the upper-half of the ellipse defined by (24) and the image of the line segment $y = -b$ is the bottom-half of the ellipse. Thus, the horizontal line segments shown in color in Figure 4.3.2(a) are mapped onto the ellipses shown in gray in Figure 4.3.2(b). The innermost pair of horizontal line segments are mapped onto the innermost ellipse, the middle pair of line segments are mapped onto the middle ellipse, and the outermost pair of line segments are mapped onto the outermost ellipse. As a final point, observe that if $b = 0$, then the image of the line segment $y = 0$, $-\pi/2 < x < \pi/2$, is the line segment $-1 \le u \le 1$, $v = 0$ on the real axis.

The mapping $w = \cos z$ can be analyzed in a similar manner, or, since $\cos z = \sin(z + \pi/2)$ from (9), we can view the mapping $w = \cos z$ as a composition of the translation $w = z + \pi/2$ and the mapping $w = \sin z$. See Problem 46 in Exercises 4.3.

4.3.2 Complex Hyperbolic Functions

A review of real hyperbolic functions can be found in Section 4.3 of the student study guide.

▶ The *real* hyperbolic sine and hyperbolic cosine functions are defined using the *real* exponential function as follows:

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

The *complex* hyperbolic sine and cosine functions are defined in an analogous manner using the *complex* exponential function.

DEFINITION 4.3.2 Complex Hyperbolic Sine and Cosine

The complex **hyperbolic sine** and **hyperbolic cosine** functions are defined by:

$$\sinh z = \frac{e^z - e^{-z}}{2} \quad \text{and} \quad \cosh z = \frac{e^z + e^{-z}}{2}. \tag{25}$$

Since the complex exponential function agrees with the real exponential function for real input, it follows from (25) that the complex hyperbolic functions agree with the real hyperbolic functions for real input. However, unlike the real hyperbolic functions whose graphs are shown in Figure 4.3.1, the complex hyperbolic functions are periodic and have infinitely many zeros. See Problem 50 in Exercises 4.3.

All zeros of $\sinh z$ and $\cosh z$ are purely imaginary.

▶ The complex hyperbolic tangent, cotangent, secant, and cosecant are defined in terms of $\sinh z$ and $\cosh z$:

$$\tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}, \quad \operatorname{sech} z = \frac{1}{\cosh z}, \quad \text{and} \quad \operatorname{csch} z = \frac{1}{\sinh z}. \tag{26}$$

Observe that the hyperbolic sine and cosine functions are entire because the functions e^z and e^{-z} are entire. Moreover, from the chain rule (6) in Section 3.2, we have:

$$\frac{d}{dz}\sinh z = \frac{d}{dz}\left(\frac{e^z - e^{-z}}{2}\right) = \frac{e^z + e^{-z}}{2}$$

or

$$\frac{d}{dz}\sinh z = \cosh z.$$

A similar computation for $\cosh z$ yields

$$\frac{d}{dz}\cosh z = \sinh z.$$

Derivatives of the remaining four remaining hyperbolic functions can then be found using (26) and the quotient rule (5) in Section 3.2.

Derivatives of Complex Hyperbolic Functions

$$\frac{d}{dz}\sinh z = \cosh z \qquad \frac{d}{dz}\cosh z = \sinh z$$
$$\frac{d}{dz}\tanh z = \operatorname{sech}^2 z \qquad \frac{d}{dz}\coth z = -\operatorname{csch}^2 z$$
$$\frac{d}{dz}\operatorname{sech} z = -\operatorname{sech} z \tanh z \qquad \frac{d}{dz}\operatorname{csch} z = -\operatorname{csch} z \coth z$$

Relation To Sine and Cosine The real trigonometric and the real hyperbolic functions share many similar properties. For example,

$$\frac{d}{dx}\sin x = \cos x \quad \text{and} \quad \frac{d}{dx}\sinh x = \cosh x.$$

Aside from the similar notational appearance and the similarities of their respective Taylor series, there is no simple way to relate the real trigonometric and the real hyperbolic functions. When dealing with the *complex* trigonometric and hyperbolic functions, however, there is a simple and beautiful connection between the two. We derive this relationship by replacing z with iz in the definition of $\sinh z$ and then comparing the result with (4):

$$\sinh(iz) = \frac{e^{iz} - e^{-iz}}{2} = i\left(\frac{e^{iz} - e^{-iz}}{2i}\right) = i\sin z,$$

or

$$-i\sinh(iz) = \sin z.$$

In a similar manner, if we substitute iz for z in the expression for $\sin z$ and compare with (25), then we find that $\sinh z = -i\sin(iz)$. After repeating this process for $\cos z$ and $\cosh z$ we obtain the following important relationships between the complex trigonometric and hyperbolic functions:

$$\sin z = -i\sinh(iz) \quad \text{and} \quad \cos z = \cosh(iz) \tag{27}$$

$$\sinh z = -i\sin(iz) \quad \text{and} \quad \cosh z = \cos(iz). \tag{28}$$

Relations between the other trigonometric and hyperbolic functions can now be derived from (27) and (28). For example,

$$\tan(iz) = \frac{\sin(iz)}{\cos(iz)} = \frac{i\sinh z}{\cosh z} = i\tanh z.$$

We can also use (27) and (28) to derive hyperbolic identities from trigonometric identities. We next list some of the more commonly used hyperbolic identities. Each of the results in (29)–(32) is identical to its real analogue.

$$\sinh(-z) = -\sinh z \quad \cosh(-z) = \cosh z \tag{29}$$

$$\cosh^2 z - \sinh^2 z = 1 \tag{30}$$

$$\sinh(z_1 \pm z_2) = \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2 \tag{31}$$

$$\cosh(z_1 \pm z_2) = \cosh z_1 \cosh z_2 \pm \sinh z_1 \sinh z_2 \tag{32}$$

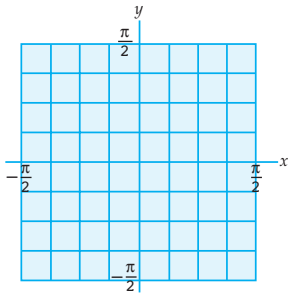
In the following example we verify the addition formula given in (32). The other identities can be verified in a similar manner. See Problems 29 and 30 in Exercises 4.3.

EXAMPLE 4 A Hyperbolic Identity

Verify that $\cosh(z_1 + z_1) = \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2$ for all complex z_1 and z_2.

Solution By (28), $\cosh(z_1 + z_2) = \cos(iz_1 + iz_2)$, and so by the trigonometric identity (9) and additional applications of (27) and (28), we obtain:

$$\begin{aligned}\cosh(z_1 + z_2) &= \cos(iz_1 + iz_2)\\ &= \cos iz_1 \cos iz_2 - \sin iz_1 \sin iz_2\\ &= \cos iz_1 \cos iz_2 + (-i\sin iz_1)(-i\sin iz_2)\\ &= \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2.\end{aligned}$$



(a) The square S_0

(b) The image E of S_0

FIGURE 4.3.3 The mapping $w = \sin z$

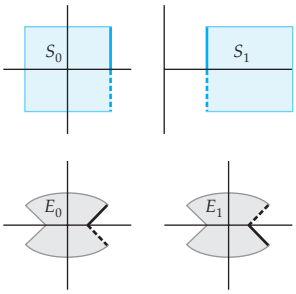


FIGURE 4.3.4 The cut elliptical regions E_0 and E_1

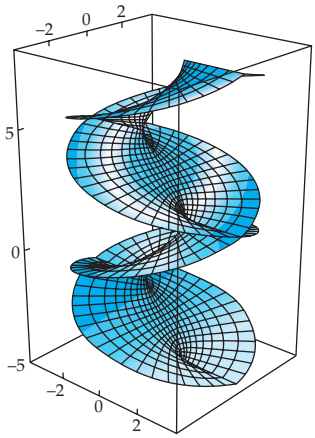


FIGURE 4.3.5 A Riemann surface for $w = \sin z$

The relations between the complex trigonometric and hyperbolic functions given in (27) and (28) also allow us determine the action of the hyperbolic functions as complex mappings. For example, because $\sinh z = -i \sin(iz)$, the complex mapping $w = \sinh z$ can be considered as the composition of the three complex mappings $w = iz$, $w = \sin z$, and $w = -iz$. See Problem 47 in Exercises 4.3.

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In real analysis, the exponential function was just one of a number of apparently equally important elementary functions. In complex analysis, however, the complex exponential function assumes a much greater role. All of the complex elementary functions can be defined solely in terms of the complex exponential and logarithmic functions. A recurring theme throughout the study of complex analysis involves using the exponential and logarithmic functions to evaluate, differentiate, integrate, and map with elementary functions.

(*ii*) As functions of a real variable x, $\sinh x$ and $\cosh x$ are not periodic. In contrast, the complex functions $\sinh z$ and $\cosh z$ are periodic. See Problem 49 in Exercises 4.3. Moreover, $\cosh x$ has no zeros and $\sinh x$ has a single zero at $x = 0$. See Figure 4.3.1. The complex functions $\sinh z$ and $\cosh z$, on the other hand, both have infinitely many zeros. See Problem 50 in Exercises 4.3.

(*iii*) Since the complex sine function is periodic, the mapping $w = \sin z$ is not one-to-one on the complex plane. Constructing a Riemann surface, for this function, as described in the *Remarks* at the end of Section 2.4 and Section 4.1, will help us visualize the complex mapping $w = \sin z$. In order to construct a Riemann surface consider the mapping on the square S_0 defined by $-\pi/2 \le x \le \pi/2$, $-\pi/2 \le y \le \pi/2$. From Example 3, we find that the square S_0 shown in color in **FIGURE 4.3.3(a)** maps onto the elliptical region E shown in gray in Figure 4.3.3(b). Similarly, the adjacent square S_1 defined by $\pi/2 \le x \le 3\pi/2$, $-\pi/2 \le y \le \pi/2$, also maps onto E.

A Riemann surface is constructed by starting with two copies of E, E_0, and E_1, representing the images of S_0 and S_1, respectively. We then cut E_0 and E_1 open along the line segments in the real axis from 1 to $\cosh(\pi/2)$ and from -1 to $-\cosh(\pi/2)$. As shown in **FIGURE 4.3.4**, the segment shown in color in the boundary of S_0 is mapped onto the segment shown in black in the boundary of E_0, while the dashed segment shown in color in the boundary of S_0 is mapped onto the dashed segment shown in black in the boundary of E_0. In a similar manner, the segments shown in color in the boundary of S_1 are mapped onto the segments shown in black in the boundary of E_1. Part of the Riemann surface consists of the two elliptical regions E_0 and E_1 with the segments shown in black glued together and the dashed segments glued together. To complete the Riemann surface, we take for every integer n an elliptical region E_n representing the image of the square S_n defined by $(2n-1)\pi/2 \le x \le (2n+1)\pi/2$, $-\pi/2 \le y \le \pi/2$. Each region E_n is cut open, as E_0 and E_1 were, and E_n is glued to E_{n+1} along their boundaries in a manner analogous to that used for E_0 and E_1. This Riemann surface, placed in xyz-space, is illustrated in **FIGURE 4.3.5**.

EXERCISES 4.3

Answers to selected odd-numbered problems begin on page ANS-13.

4.3.1 Complex Trigonometric Functions

In Problems 1–8, express the value of the given trigonometric function in the form $a + ib$.

1. $\sin(4i)$

2. $\cos(-3i)$

3. $\cos(2 - 4i)$

4. $\sin\left(\frac{\pi}{4} + i\right)$

5. $\tan(2i)$

6. $\cot(\pi + 2i)$

7. $\sec\left(\frac{\pi}{2} - i\right)$

8. $\csc(1 + i)$

In Problems 9–12, find all complex values z satisfying the given equation.

9. $\sin z = i$

10. $\cos z = 4$

11. $\sin z = \cos z$

12. $\cos z = i \sin z$

In Problems 13–16, verify the given trigonometric identity.

13. $\sin(-z) = -\sin z$

14. $\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$

15. $\overline{\cos z} = \cos \bar{z}$

16. $\sin\left(z - \frac{\pi}{2}\right) = -\cos z$

In Problems 17–20, find the derivative of the given function.

17. $\sin(z^2)$

18. $\cos(ie^z)$

19. $z \tan \frac{1}{z}$

20. $\sec\left(z^2 + (1 - i)z + i\right)$

4.3.2 Complex Hyperbolic Functions

In Problems 21–24, express the value of the given hyperbolic function in the form $a + ib$.

21. $\cosh(\pi i)$

22. $\sinh\left(\frac{\pi}{2} i\right)$

23. $\cosh\left(1 + \frac{\pi}{6} i\right)$

24. $\tanh(2 + 3i)$

In Problems 25–28, find all complex values z satisfying the given equation.

25. $\cosh z = i$

26. $\sinh z = -1$

27. $\sinh z = \cosh z$

28. $\sinh z = e^z$

In Problems 29–32, verify the given hyperbolic identity.

29. $\cosh^2 z - \sinh^2 z = 1$

30. $\sinh(z_1 + z_2) = \sinh z_1 \cosh z_2 + \cosh z_1 \sinh z_2$

31. $|\sinh z|^2 = \sinh^2 x + \sin^2 y$

32. $\operatorname{Im}(\cosh z) = \sinh x \sin y$

In Problems 33–36, find the derivative of the given function.

33. $\sin z \sinh z$

34. $\tanh z$

35. $\tanh(iz - 2)$

36. $\cosh\left(iz + e^{iz}\right)$